PhytoPi

PHYTOPI

PROJECT CHARTER

VERSION <0.1>

<09/28/2025>

REVISION HISTORY

Date	Version	Description	Author
09/28/25	*0.1*	*First draft of project charter*	*N. Tuttle*

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PURPOSE/PROJECT JUSTIFICATION

The PhytoPi project was created to address the common issue among gardening hobbyists. Traditional plant maintenance requires regular watering and diagnosis of health issues. PhytoPi was designed to reduce these burdens with an automated, IoT-based solution that minimizes need for human intervention and ensures optimal growth.

The primary goal of PhytoPi is to create a highly affordable, automated plant care system that integrates both embedded hardware and software solutions. This project requires data collection using sensors and a camera, which also serves our requirement that the product should analyze collected data for health trends of the plant. This project also must have a trigger system so that the plant can be watered, humidity adjusted, automatically to avoid human intervention. PhytoPi requires a mobile interface that can deliver alerts about plant status to users with a remote dashboard for precise control over the device. There is an increasing demand for urban agriculture and remote plant care, and we aim to create a product that stands out against plant growth monitor competitors by offering ease-of-use for end users, detailed statistics about plant status, and utilizes efficient low-level hardware that can interface with readily available components for customer appeal and low costs.

OBJECTIVES AND SUCCESS CRITERIA

Business Strategy objective	Project objective
To improve urban agriculture and promote sustainability.	To reduce manual plant care labor by at least 80%.
To compete for a share of the market in IoT solutions through affordable components.	To deliver a product with hardware costs under $250/unit, target costs for prototype production is $200.
To create a user interface that keeps customers extremely engaged through speed/ease of use.	To create a responsive mobile dashboard that has a refresh time of less than 2 seconds
To create a product that collects precise details about plant growth for data insights.	To collect and analyze at least 3 different plant health metrics over the life cycle of a plant including light levels, soil moisture, and projected growth vs. actual growth.

REQUIREMENTS

1. Core Features:
    a. The system must monitor soil moisture, light, temperature, and humidity levels
    b. The system must respond to sensor readings and adjust environmental factors accordingly such as watering the plant for soil moisture.
    c. The system must store plant health data collected from sensors and cameras.
    d. The system must analyze stored data about a plant's growth to identify trends in health and potential issues

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    e. The prototyped system will support one plant species, scaling for multiple species and further training of the machine learning model will be explored after initial prototype.
2. Hardware Requirements:
    a. The system must be built using a Raspberry Pi 5 used to read connected sensors (soil moisture, humidity, etc.)
    b. The system must have a pump that effectively transfers water to the plant's soil for automatic irrigation
    c. The system must be housed in a suitable enclosure that can be used near plants and water sources.
3. User Experience:
    a. The user must have a mobile interface that provides real-time data/status of the plant
    b. The system must alert/notify the user of any unexpected plant conditions.
    c. Users should be able to adjust settings for the mobile interface and the PhytoPi system remotely
4. Performance:
    a. After a correct initial setup, the system should operate continuously with minimal user intervention.
    b. The initial prototype must support one plant throughout its entire life cycle; scaling may be explored in future iterations.
    c. Sensor readings and general performance should be extremely responsive and sensor interfacing code should be written in C where possible.

PROJECT ASSUMPTIONS AND CONSTRAINTS

- Constraints:
    - Hardware: Project must utilize a Raspberry Pi 5 and an ESP32, even if production design may be scaled down to smaller hardware after prototyping.
    - Scope: Prototype version will support one plant species at a time (species defined in requirements section)
    - Timeline: Must be completed before the end of the current academic term
    - Budget: Low-cost off-the-shelf components must be used, funding is not available for large-scale manufacturing, budget limited to $300 per prototype maximum
    - Environment: System must be able to function safely and effectively in an environment near plants and water.
- Assumptions:
    - Sensors: Readily available sensors being used in the project will provide accurate and quick readings suitable for providing accurate data analysis and for adjusting environmental factors to improve plant conditions.
    - Users: Users will have a smartphone or similar device for using the mobile dashboard and adjusting settings.
    - Connectivity: Users will have access to a Wi-Fi connection for connecting their mobile dashboard to the PhytoPi.

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- ML Implementation: Machine learning models will be capable of analyzing plant health accurately using a camera while running locally on a Raspberry Pi 5.
- Market: The increasing interest in urban agriculture and automated plant care will remain or increase.

PRELIMINARY RISK STATEMENT

Date raised	Risk description	Likelihood	Impact	Risk Rating	Mitigating action	Contingent action
9/26/25	There is a risk that if the ML model or sensors misidentify plant health issues, there may be incorrect environmental adjustments.	Medium	High	High	Limit prototype to one plant species, combine sensor thresholds with ML outputs to ensure ML does not over adjust incorrectly. Use reliable, proven off-the-shelf sensors, calibrate sensors before deployment.	Send alert to user if ML confidence is low, user can adjust camera or lighting if needed. If sensor readings are unexpected, push alerts to user through mobile dashboard, ignore extreme outliers.
9/26/25	There is a risk that the Raspberry Pi 5, ESP32, or pump motors fails and halts automated operation	Low	High	Medium	Test all hardware before integration, be ready with spare components and use a good power source.	Halt automated actions, notify user, replace failed hardware.
9/26/25	There is a risk that there will be timeline delays, risking project incompletion before the term ends.	Medium	High	High	Break project into milestones weekly/biweekly, monitor tasks using project management software.	Reprioritize if a milestone is not accomplished in time, extend working hours, communicate delays through messaging software.

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9/26/25	There is a risk that the budget will overrun and potentially not meet our target cost of $200 or exceed our maximum of $300.	Low	Low	Low	Track costs when purchasing components, compare multiple suppliers and prioritize core functionality components.	Remove optional components, adjust specifications to remain within budget, 3D print parts such as the plant enclosure.
9/26/25	There is a risk that users will damage components of the device with a water source or mishandling.	Low	High	Medium	Include clear instructions for users, design protective enclosure and waterproof sensitive electronics.	Replace affected component, provide troubleshooting steps for users.
9/26/25	There is a risk that the plant species chosen for prototyping will not be able to survive in the PhytoPi's controlled environment	Medium	Medium	Medium	Choose low-maintenance, hardy plant species, control environment carefully.	Replace with a new specimen, adjust environmental factors and retry with different factors.
9/26/25	There is a risk that user Wi-Fi connectivity will drop, preventing remote control and monitoring.	Low	Low	Low	Test on typical network setups, implement manual local control, add reconnection logic.	If Wi-Fi is down, continue automated functions normally, alert user if connection reestablished.

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SUMMARY MILESTONE SCHEDULE

Project Milestone	**Timeline**	**Owner**
[Initiation]	Sep 2025	Daniel Grijalva
[Concept Development]	Sep 2025	Nolan Tuttle
[Planning]	Oct 2025	Full Team
[Requirements Analysis]	Oct 2025	Full Team
[Implementation Phase 1 (Cloud + App)]	Nov-Feb 2026	Daniel Grijalva
[Implementation Phase 2 (Hardware)]	Feb-Mar 2026	Nolan Tuttle
Testing/Integration	Mar 2026	Full Team
Finalization and Showcase Prep	Apr 2026	Full Team

SUMMARY ESTIMATED BUDGET

The forecasted budget for the PhytoPi project is $700. It is to be funded through personal funds and GCU's capstone fund. Two prototypes will be developed, thus making the ROM estimate for Hardware the cost of both prototypes.

Rough Order of Magnitude Item	**ROM Estimate**
[Software]	$0
[Internal Resources]	$0
[Contractor Resources]	$0
[Hardware/Physical Resources]	$600-$700

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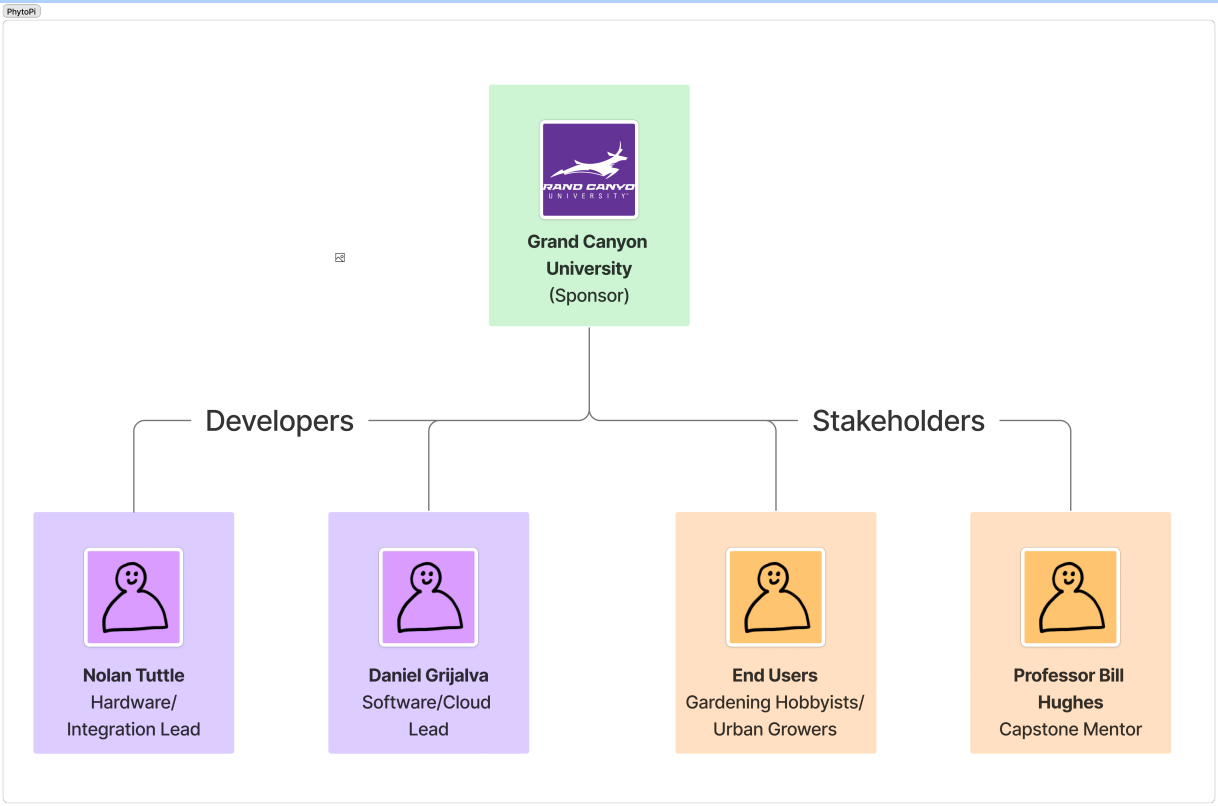
PROJECT ORGANIZATION

PROJECT ORGANIZATION CHART



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ROLES AND RESPONSIBILITIES

RESPONSIBILITY ASSIGNMENT MATRIX

	Nolan Tuttle		Daniel Grijalva		Professor Bill Hughes
Initiating Project	*R*	*A*	*R*	*C*	*I*
Defining Requirements	*R*	*C*	*R*	*A*	*I*
Mobile App Development	*R*	*C*	*R*	*A*	*I*
Database Configuration	R	C	R	A	I
C Programming for Hardware Interfacing	R	A	R	C	C
Developing Enclosure and System Design	R	A	R	C	C
Testing	R	A	R	C	C
Deployment	R	C	R	A	C
Documentation	R	A	R	A	I

PROJECT GOVERNANCE REQUIREMENTS

1. Decision-Making
    a. Sponsor (GCU): Final approval on project scope, deliverables, and presentation.
    b. Project Team: Responsible for day-to-day decisions on project design and development.
    c. Capstone Mentor: Consultant for technical decisions, help to make project compliant with university standards.
2. Check-ins
    a. Weekly check-in with capstone class, stand-up status updates.
    b. Weekly check-in with capstone mentor, updates on progress, consulting for design choices, scope, or documentation for project.
    c. Weekly syncs between team members, in-person or video call.

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3. Changes/QA
    a. Both team members/developers share planning, testing, documentation, and presentation.
    b. Shared GitHub repository with branching for smooth integration.
    c. Peer review between team members for new code, hardware integration, and documentation.
    d. All changes logged in Jira, team review over timeline/impact. Advisor approval required for scope/time changes.
4. Risk/Issue Management
    a. Risks identified and tracked using risk register.
    b. Mentor consulting for high-impact risks, mitigation strategies implemented.

PROJECT MANAGER RESPONSIBILITIES

1. Planning/Scheduling
    a. Developing and maintaining a project timeline, split tasks into milestones/sprints
    b. Ensure tasks are completed by assigned deadlines, assign responsibility
2. Scope
    a. Define project scope and expected functionality with mentor and team
    b. Ensure scope changes are documented and approved by mentor
3. Resource/Risk Management
    a. Identify all hardware/software resources required for development
    b. Maintain the risk register and escalate issues in a timely manner as needed
4. Communication
    a. Primary point of contact with capstone mentor
    b. Responsible for communicating status updates, progress reports, risks and issues.

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